

Online instantaneous stress detection from physiological and physical markers in shooting simulation by Multi-Dimensional Scaling

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ENEE 691: Signal Processing Approaches for Big Data Analytics

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Introduction: Detecting or predicting stress in day-to-day scenario is becoming attainable and ubiquitous due to the enormous data which can be extracted from smartphones [1] and other low-cost sensors [2]. But detecting stress in flight or fight situations or in very small duration of time is still an open research topic because of the detecting complications of stress. In this project, we tried to predict stress using online methods using two approaches:

1. Classical machine learning approach by finding key features (Model driven approach)
2. Multi-Dimensional Scaling approach (Data driven approach)

We compared these two approaches by finding the accuracy and execution time. The classical machine learning approach yields a best-case accuracy of 89.58% with execution time of 7.08 seconds. On the contrary Multi-Dimensional Scaling (MDS) approach yields a best-case accuracy of 84.31% with execution time of 1.16 seconds which yields 83.61% performance increase with a drop of only 5.27% classification accuracy. Which proves that MDS approach is best suited for this application in case of online prediction.

Problem statement: Instantaneous stress detection is a process of stress prediction using data of a very small period of time Δt (typically $\Delta t < 10$ seconds). In Eclipse lab, we collected data of various sensors (Accelerometer, Heart rate, Blood pressure, Galvanic skin response) during a shooting simulation game for 17 subjects. The game session was consisted of 6 sessions each with a low stress and high stress condition. In low stress condition, the shooting target appears slowly and for high stress condition, it appears faster. In this project, we sought to predict the stress of the subject in between the time when the target is visible in the simulation screen. Figure 1 and 2 depicts the shooting screen of the simulator and the subject respectively.



Figure 1: Shooting simulation screen



Figure 2: Subject

Prior art: Similar work of prediction of stress was performed by Dobbins et al [3] but they implemented the whole system for driving scenarios. Moreover, Castaldo et al [4] claims that for ultra-short period, HRV (Heart rate variability) is a significant physiological marker for stress.

Problem formulation: Detecting stress is complicated because of the inter-subject variation of physiological and physical data. Moreover, in ultra-short period ($\Delta t < 10$ seconds) detection of stress becomes more tedious because the physiological markers (Heart rate, Galvanic skin response) doesn't change fast enough. That's why detecting instantaneous stress is nearly impossible with only physiological markers. To counter that problem, we added some physical markers (Gun velocity, Accelerometer) for compensation. Furthermore, to make the algorithm online, we need to make sure the algorithm is fast enough to classify within 2 seconds. Multi-dimensional scaling technique helps a lot in this case because it finds the Euclidean distance within the data points and provides us a lower dimensional representation of data. Which in turn helps to acquire our desired speed of the algorithm.

Methods: The simplified block diagram of classical machine learning approach taken for this project is presented in Figure 3. The details of the blocks will be presented in Experimental setup section.

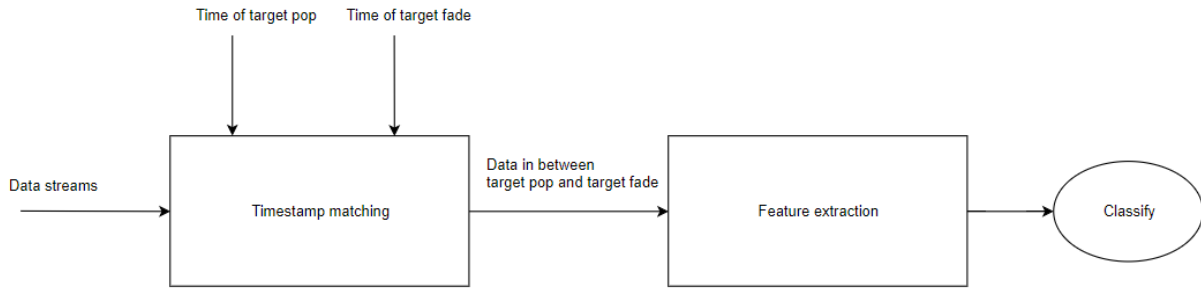


Figure 3: Block diagram of classical machine learning approach

The block diagram of MDS approach is somewhat similar with the classical machine learning approach, but we employed a Least Square or Kruskal-Shephard Multi-dimensional scaling (LS-MDS) because it preserves the dissimilarity between data points unlike Classical MDS. The cost function of LS-MDS is presented in Equation 1. The goal is to minimize the cost function along $y_1 \dots y_N$.

$$\min_{Y \in \mathbb{R}^{N \times q}} \sum_{i < j} w_{ij} \left(\delta_{ij} - d_{ij}(Y) \right)^2 \quad (1)$$

Where,

$$d_{ij}(Y) = \|y_i - y_j\|_2$$

$$w_{ij} = \text{Weights}$$

The block diagram is provided in Figure 4. More details of the blocks are presented in Experimental setup section.

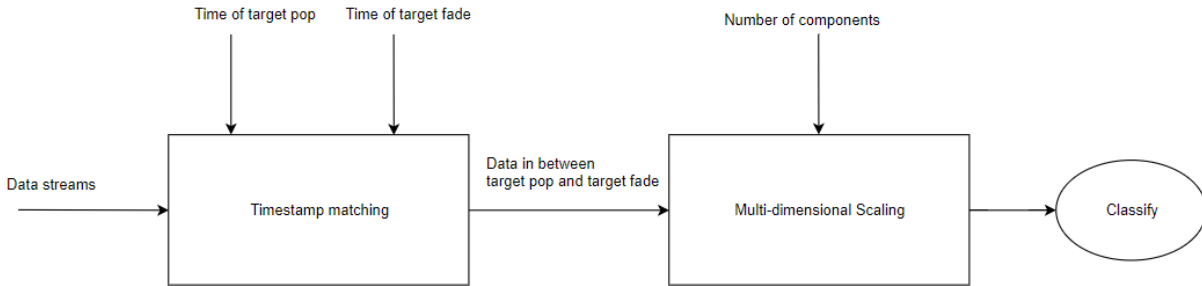


Figure 4: Block diagram of Multi-dimensional scaling approach

Experimental setup:

- Timestamp matching:** As we are using data from different sensors, we need to match the timestamps to maintain the temporal significance. We extracted the timestamp when the target popped up in the screen (Target pop time) and the timestamp when the target faded (Target fade time). Then a timestamp matching algorithm depicted in Figure 5 was applied to extract the sensor data in between target pop and target fade. This step was done because the motivation of this project is to predict if the subject was stressed or not when the target was on the screen. In Figure 6 the extraction process of the data streams and data length is provided. In total, data of 4 subjects were extracted because only their data was complete and usable.

```

out = []
index_match = []
l = len(data)
for i in target_time:
    i_array = np.repeat(i,l)
    diff = time - i_array
    matched_index = np.argmin(abs(diff))
    out.append(data[matched_index])
    index_match.append(matched_index)
return out,index_match
  
```

Figure 5: Timestamp matching algorithm

```

Channel name = EmpaticaAcceleration Data length = 63312
Channel name = EmpaticaBVP Data length = 126632
Channel name = EmpaticaGSR Data length = 7914
Channel name = EmpaticaTemperature Data length = 7912
Session 1 data written

Channel name = EmpaticaHR Data length = 2856
Session 1 data written
Channel name = EmpaticaHR Data length = 2613
Session 2 data written
Channel name = EmpaticaHR Data length = 2456
Session 3 data written
  
```

Figure 6: Snapshot of data extraction

- A. **Classical machine learning approach:** For machine learning approach 10 temporal and spectral features were extracted from each data-stream. The features are presented below.

- Temporal features (Time domain)
 - Mean
 - Standard deviation
 - Maximum value
 - L2 norm
 - Minimum value
 - Product of maximum and minimum value

- Spectral features (Frequency domain)
 1. LF peak (Peak frequency of Low-frequency band)
 2. LF power (Absolute power of Low-frequency band)
 3. HF peak (Peak frequency of High-frequency band)
 4. HF power (Absolute power of Low-frequency band)

B. **Multi-dimensional scaling approach:** The main problem with a model-based approach like key feature extraction is, it's not generalized and it might take a long extraction time. For this reason, a data-driven approach like Multi-dimensional scaling approach is suitable to address these two issues. For this project, we used Least Square or Kruskal-Shephard Multi-dimensional scaling to reduce the dimension and to preserve the relative distance of the data points. Which in terms helps to make the algorithm faster. We computed the performance of LS-MDS with various number of components and compared with the classical machine learning approach which is presented in the results section. The main reason of using LS-MDS is, preserves the dissimilarity between data points and computationally efficient which is suited for this application.

- **Classification:** For the classification, we used 3 different machine learning algorithms to compare their performances. The used classifiers are
 1. Multi-layer perceptron (MLP)
 2. Support vector machine (SVM)
 3. Logistic regression (LR)

For MLP we used a 4-layer network with 100 nodes in the first layer and 10 nodes in 2nd, 3rd and 4th layer. For the optimization, Adam optimizer and an adaptive learning rate was used.

For SVM we also used an adaptive learning rate with a rbf (Radial basis function) kernel and for hyperparameter C was chosen to be equal to 0.001. Finally, for Logistic regression, a learning rate of 0.01 was used and as the optimizer Adam optimizer was used because it is computationally efficient.

Results and discussion:

- **Classical machine learning approach:** By extracting the data of subject 28, we filtered the data of total 6 sessions by a low pass Butterworth filter to remove noise and outliers with a cutoff of 50Hz. Then we extracted the aforementioned features from selected data streams. The selected data streams are Gun velocity (GV), Blood volume pulse (BVP), Heart rate (HR), Accelerometer (Acc) and Galvanic skin response (GSR). The details of the results are presented in Table 1. The units for accuracy and the execution times are percentage (%) and seconds (s) respectively.

Features	Number of features	Accuracy (%)	Execution time (Classification)	Execution time (Extraction)
GV,BVP,HR,Acc,GSR	50	89.58	7.08	62.73
BVP,HR,Acc,GSR	40	87.361	6.26	59.77
BVP,HR,Acc	30	85.13	5.57	57.38
BVP,HR	20	79.16	3.6	53.82
HR	10	78.88	3.077	49.16

Table 1: Machine learning approach with different feature length

Then, in Figure 7 the accuracy of MLP, LR and SVM are presented for subject 21,28,29 and 31. In each case, 5 sessions data were used as training data and 1 session were kept as test data.

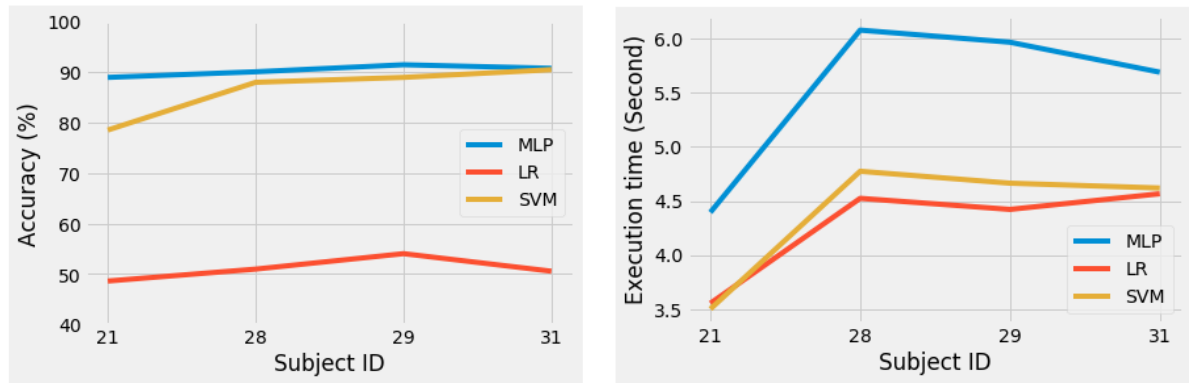


Figure 7: Accuracy and execution time for MLP, LR and SVM

From Figure 7 it is clear that, for the classification process, MLP is better than LR and SVM in terms of accuracy. But in terms of execution time, SVM performs better than MLP because it's faster.

- **Multi-dimensional scaling approach:** From the previous section, it is evident that MLP is the best classifier to use but the downfall of the classical machine learning approach is it's not fast. For online detection of stress, we need to classify the data within 2 seconds. So, to reduce the dimension of the data and induce sparsity without removing the valuable information in the data, Least Square Multi-dimensional scaling was performed. The detailed result using the data of Subject 28 is presented in Table 2

Number of components	Accuracy	Execution time (Classification)	Execution time (Scaling)
10	81.82	0.726	46.11
20	83.6	1.04	47.91
30	84.31	1.16	48.66
40	82.79	1.24	50.52
50	82.40	1.33	51.6

Table 2: LS-MDS approach for different number of components

From Table 2, it is evident that the number of components = 30, MDS provides the best performance in terms of accuracy and execution time.

- **Comparison of two approach:** In Figure 8, the comparison between MLP and MDS with 30 components for the 4 subjects are presented. In each case, 5 sessions data were used as training data and 1 session were kept as test data.

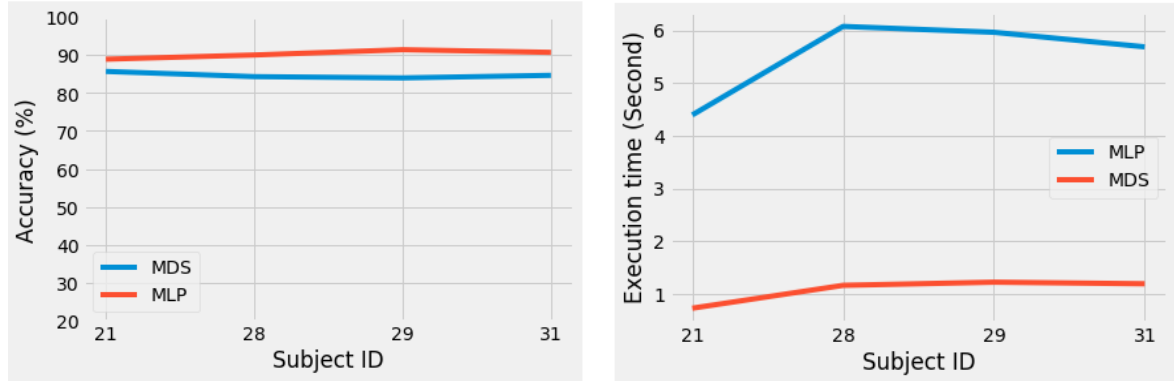


Figure 8: Comparison of MDS and MLP in terms of accuracy and execution time

From Figure 8, we can say that, the MDS approach is suitable for online detection of stress because it's much faster than MLP. On average, the execution time of MDS is 83.61% faster than MLP with a drop of only 5.27% accuracy. The main reasons behind the sluggishness classical machine learning algorithms are, it needs scale the data before classification which can be computationally expensive. Moreover, feature extraction with temporal and spectral features for large data can also be computationally expensive whereas, in MDS approach the main computationally expensive part is the cost minimization.

- **Comparison with prior art:** In case of stress detection, state of art stress detector can attain accuracy up to 96%. But in those cases, data for long period of time (generally > 5 minutes), context from the subject and time of day also taken in account. That's why in this case, those algorithms can't be applicable for comparison with out approach. Because without context of subject, time of day and large data can not be used in this case of online stress detection. In Table 3, some stress detection algorithms which specializes in online detection are compared with this approach to figure out the viability of MDS approach in stress detection.

Algorithm	Accuracy (%)	Comment
MDS approach	84.31	Faster algorithm for online classification
Wijsman at el [5]	80	Used only physiological data
Munla at el [6]	83	Used both spectral and temporal features of data
Ciabattioni at el [7]	84.5	Used data from smartwatch for long period of time (10 minutes)

Table 3: Comparison of MDS approach with prior art

Conclusion: In this project, a model-driven approach (classical machine learning) and a data-driven approach (MDS) were compared using real world human data. In case of online classification, we have found that MDS performs better than any model-driven approach because of the capability of MDS to retain information within a smaller dimension, which in terms makes the classification faster than other methods. Multiple simulations were conducted to find the optimal classifier and tune the parameters for the classifier and MDS algorithm.

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